HEARTPULSE AI

HEART RATE & FITNESS ANALYTICS

A personal exploratory data analysis project

Created by
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Python • Pandas • NumPy • Matplotlib • Seaborn

PROJECT REPORT | 2026

01 | Project Overview

A concise overview of the project, its purpose and the scope of the analysis.

Abstract. HeartPulse AI is a personal data science project exploring a tabular fitness dataset containing age, sleep duration, weekly exercise frequency and heart-rate measurements. The work uses Python-based exploratory data analysis (EDA) to inspect data quality, summarize distributions, compare groups, engineer a derived heart-rate difference feature and visualize relationships among variables. The project is intended as a learning and portfolio exercise; it is not a clinical tool and does not diagnose, assess or predict an individual's health.

Project goal. Practice a complete beginner-friendly analytics workflow: load data, inspect and clean it, calculate descriptive statistics, create meaningful visualizations, interpret patterns cautiously and prepare a reproducible dataset for further work.

Scope. This version is descriptive EDA only. No machine-learning model has been trained, no predictions are produced, and no causal conclusions are drawn.

Dataset snapshot

500	6	0	0
records	original variables	missing cells	duplicate rows

The uploaded CSV contains 500 rows and six original numeric columns. The initial quality check found no missing values and no exact duplicate rows. The analysis also creates derived columns for heart-rate difference, age category and sleep category.

02 | Questions & Dataset

The analysis is organized around practical exploratory questions:

- How do the average resting and maximum exercise heart-rate measurements compare?
- How do maximum exercise heart-rate averages vary across age groups?
- How do they vary across weekly exercise-frequency and sleep-duration groups?
- What is the distribution of the difference between the two resting heart-rate fields?
- Which variable pairs show stronger or weaker linear associations in this dataset?

Dataset fields

Original column	Meaning	Type / unit
Age	Participant age	Years
Sleep Hours	Reported sleep duration	Hours
Exercise Frequency (Days/Week)	Exercise frequency	Days per week
Resting Heart Rate Before	Resting heart rate before exercise	Beats per minute (BPM)
Resting Heart Rate After	Resting heart rate after exercise	BPM
Max Heart Rate During Exercise	Maximum heart rate recorded during exercise	BPM

Source note. The working file used for this report was named *heart_rate_data.csv*. The original public dataset URL and its collection methodology were not included in the supplied project materials. Add the verified Kaggle dataset URL and any available licensing / attribution details to the repository before publishing.

03 | Tools & Methodology

Tool	Role in this project
Python	Core programming language for the analysis.
Google Colab	Notebook environment used to run and document code.
Pandas	CSV loading, inspection, grouping, summary statistics and export.
NumPy	Numerical operations and bin definitions.
Matplotlib	Charts including bar charts, line plots and histograms.
Seaborn	Correlation heatmap visualization.

Analysis workflow

1. Load and inspect: read the CSV into a Pandas DataFrame and inspect sample rows, column names, data types and descriptive statistics.
2. Assess data quality: check missing values and exact duplicate rows. The supplied file contained zero missing cells and zero duplicate rows, so no rows were removed in this report's reproduction.
3. Explore distributions: calculate means, medians, minima, maxima and standard deviations; create histograms and a box plot for numerical spread and potential outliers.
4. Compare groups: calculate average maximum exercise heart rate by age category, sleep category and exercise frequency.
5. Engineer a feature: calculate *Heart Rate Difference = Resting Heart Rate After – Resting Heart Rate Before*.
6. Examine associations: calculate Pearson correlations among numeric variables and display them in a heatmap.
7. Export: save a processed CSV containing the original fields and derived analysis columns.

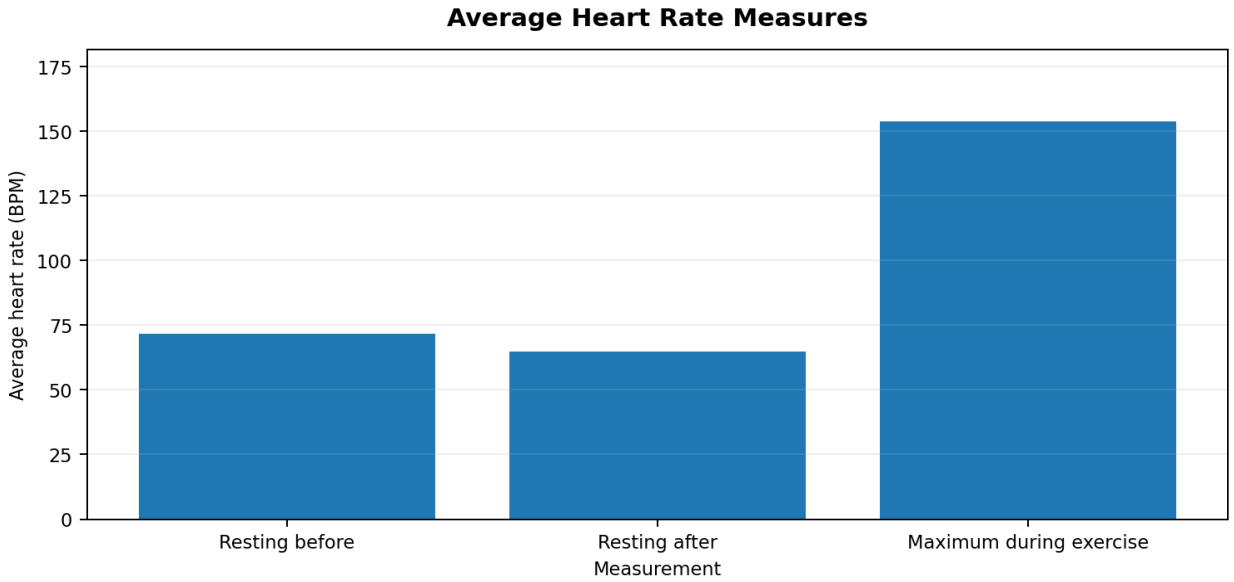
Reproducibility note

The figures and statistics in this report were recalculated directly from the supplied CSV. Derived category columns use the same group boundaries described in the notebook. Minor differences may occur if the source CSV is later updated.

04 | Descriptive Statistics

Variable	Mean	Median	Min	Max	Std. dev.
Age	40.03	41.00	18.00	60.00	12.18
Sleep Hours	6.99	7.00	2.90	11.40	1.43
Exercise days/week	3.96	4.00	1.00	7.00	2.08
Resting HR before	71.64	72.00	45.00	100.00	8.37
Resting HR after	64.68	65.00	41.00	93.00	8.57
Maximum exercise HR	153.89	154.00	124.00	178.00	9.17
HR difference	-6.96	-7.00	-10.00	-4.00	1.93

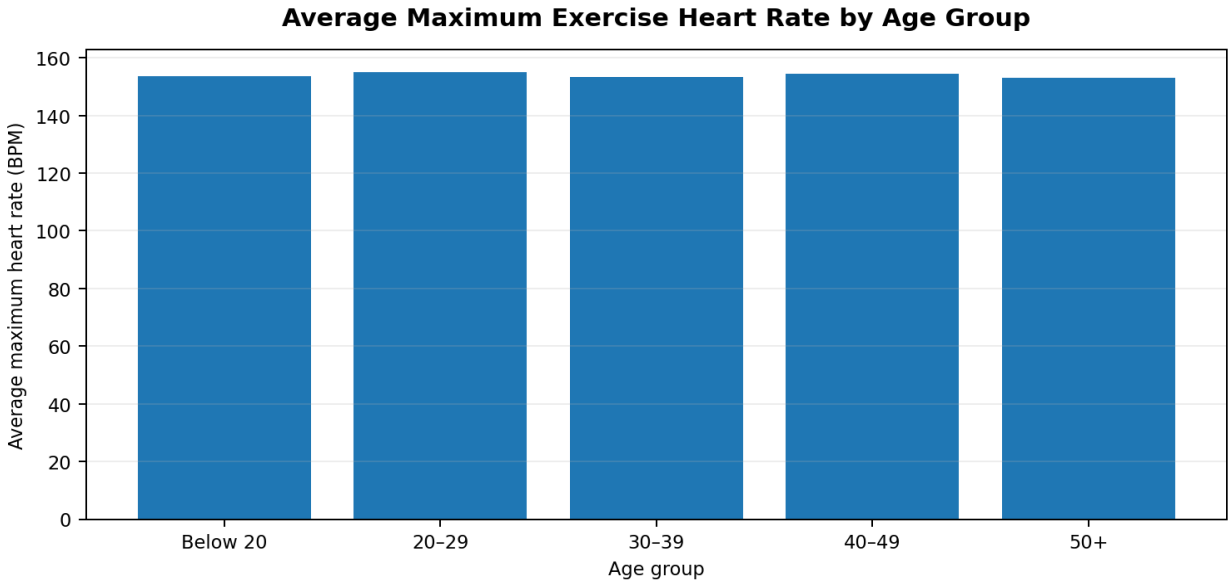
Selected headline averages



The average resting heart rate before exercise was 71.64 BPM; the average value in the field labelled after exercise was 64.68 BPM; and the average maximum heart rate during exercise was 153.89 BPM. These are dataset-level averages, not individual targets or clinical reference judgments.

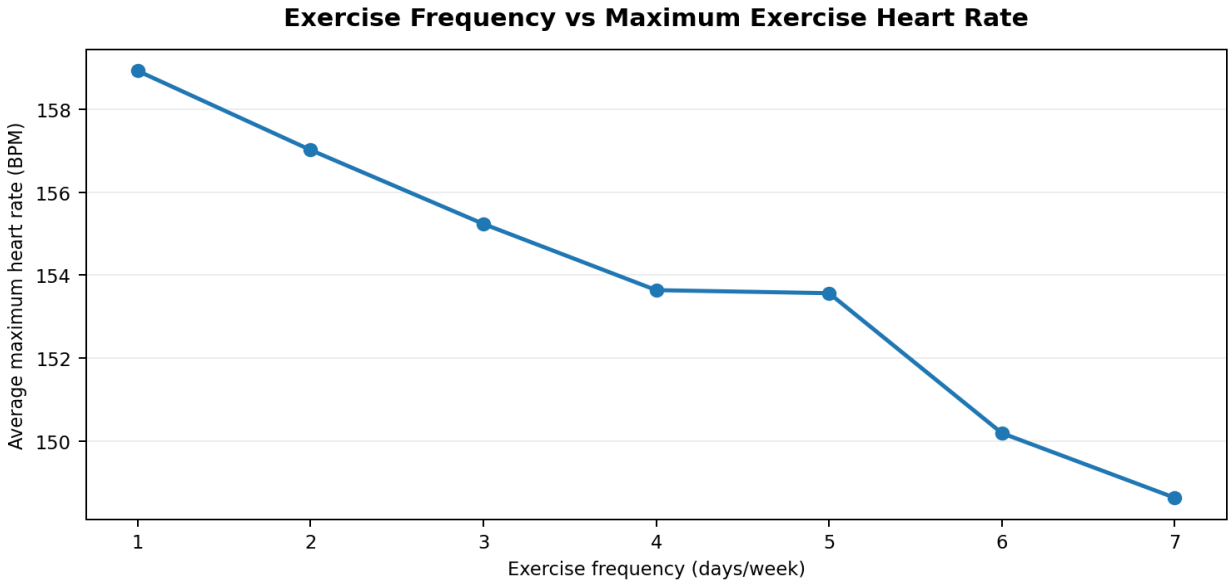
05 | Group Comparisons

Age groups



The average maximum exercise heart rate varied modestly across the age bands. The group averages should be interpreted with their group sizes in mind and should not be treated as age-specific physiological norms.

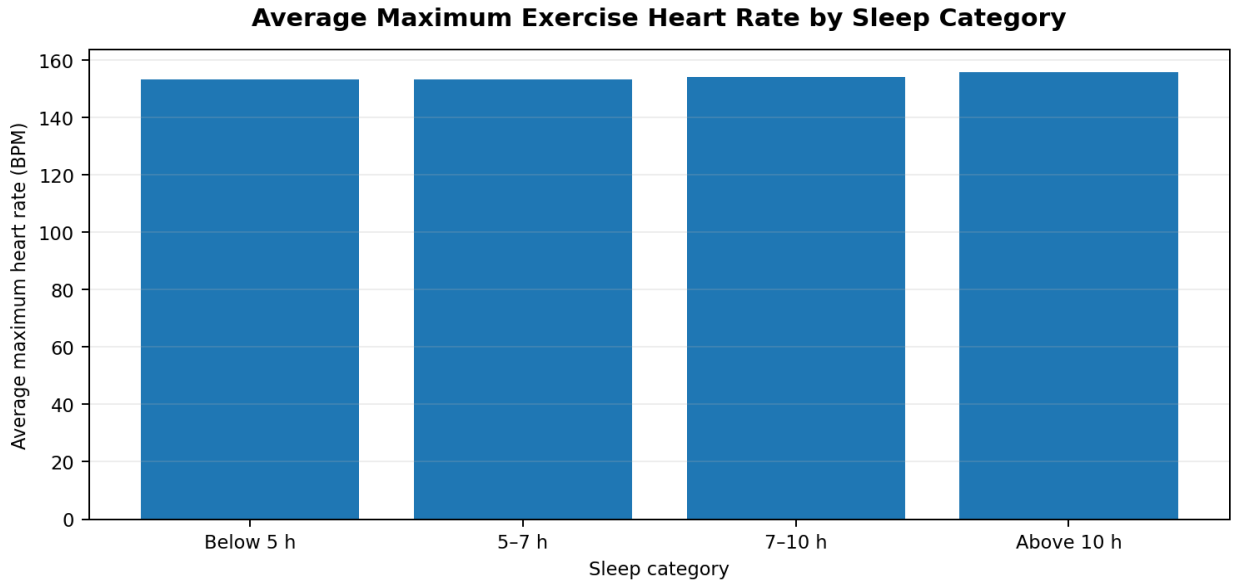
Weekly exercise frequency



In this dataset, the average maximum exercise heart-rate value generally decreased across the higher weekly exercise-frequency groups. This is an observed association in this particular dataset; it does not establish that exercising more often causes a lower maximum heart rate.

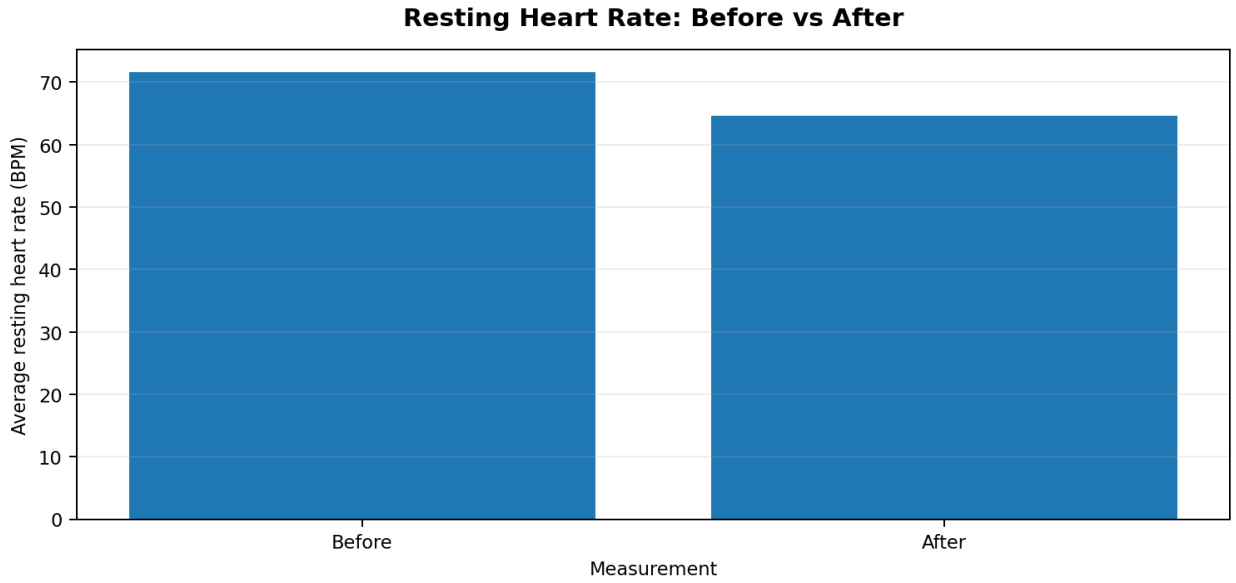
06 | Sleep, Resting Heart Rate & Feature Engineering

Sleep-duration categories



The mean maximum exercise heart-rate values are relatively close across the sleep groups. The group above 10 hours contains only 11 records, so its average is less stable and should be treated cautiously.

Resting heart rate before and after

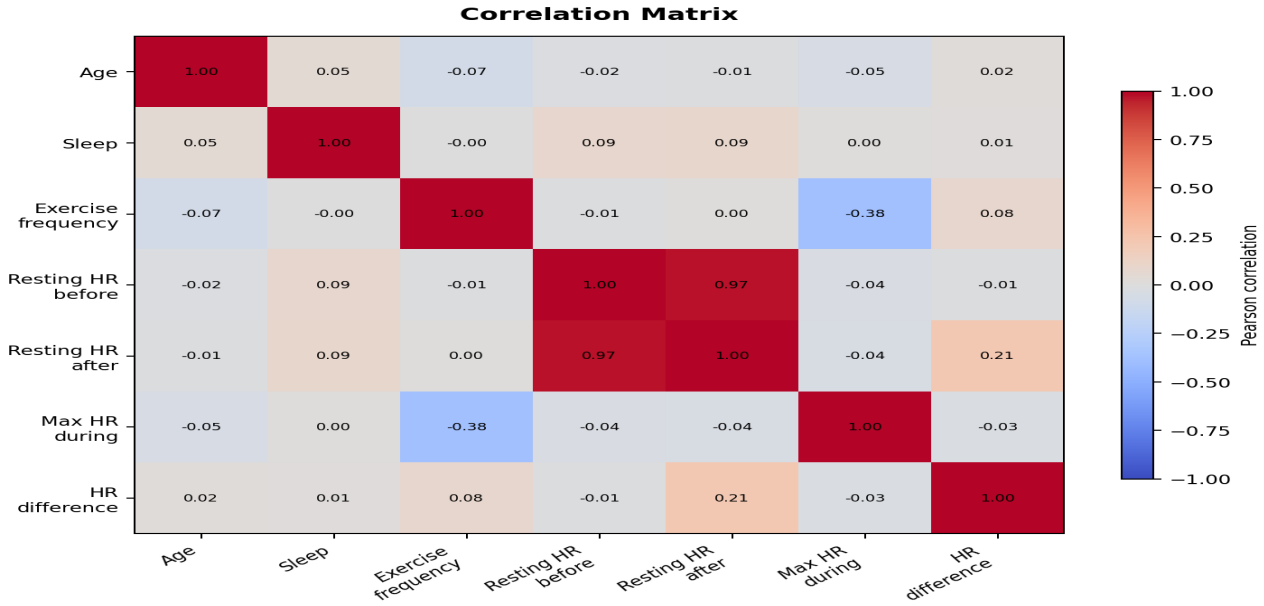


The mean of the field labelled 'Resting Heart Rate After' is 6.96 BPM lower than the mean of 'Resting Heart Rate Before'. The dataset documentation provided for this report does not specify the exact timing or measurement protocol for the 'after' reading, so this difference must not be described as a validated heart-rate recovery measure.

Derived feature: Heart Rate Difference

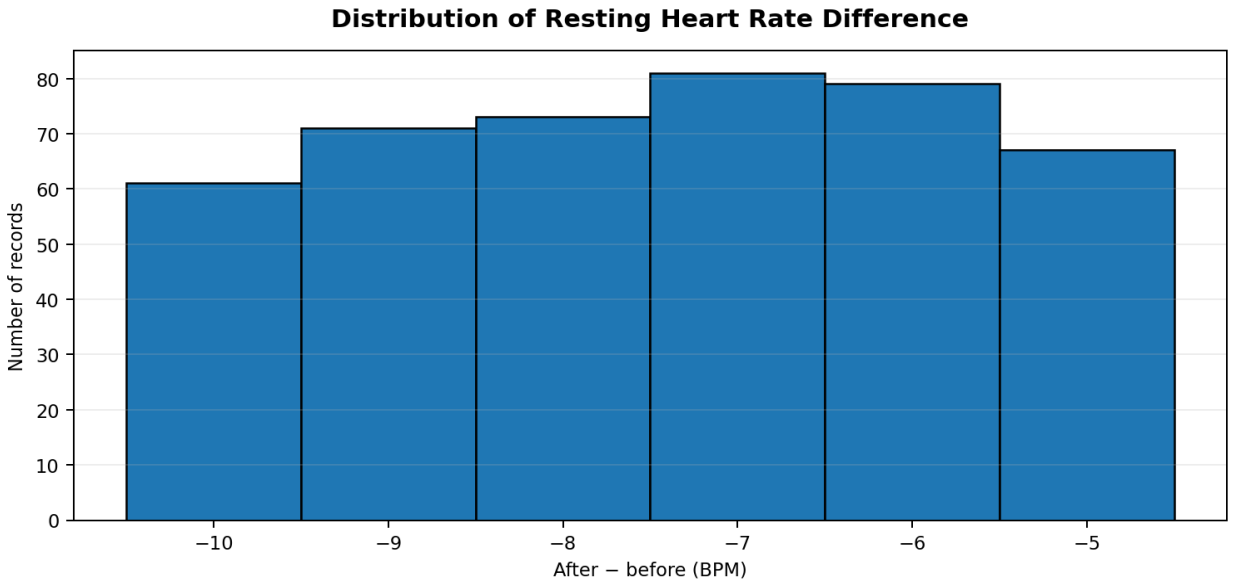
A new feature was calculated as **after** – **before**. Its mean is –6.96 BPM, median is –7 BPM, and observed range is –10 to –4 BPM. Negative values mean the recorded 'after' value is lower than the corresponding 'before' value. The feature is useful for practicing feature engineering, but its physiological interpretation depends on the missing measurement protocol.

07 | Correlation & Distribution

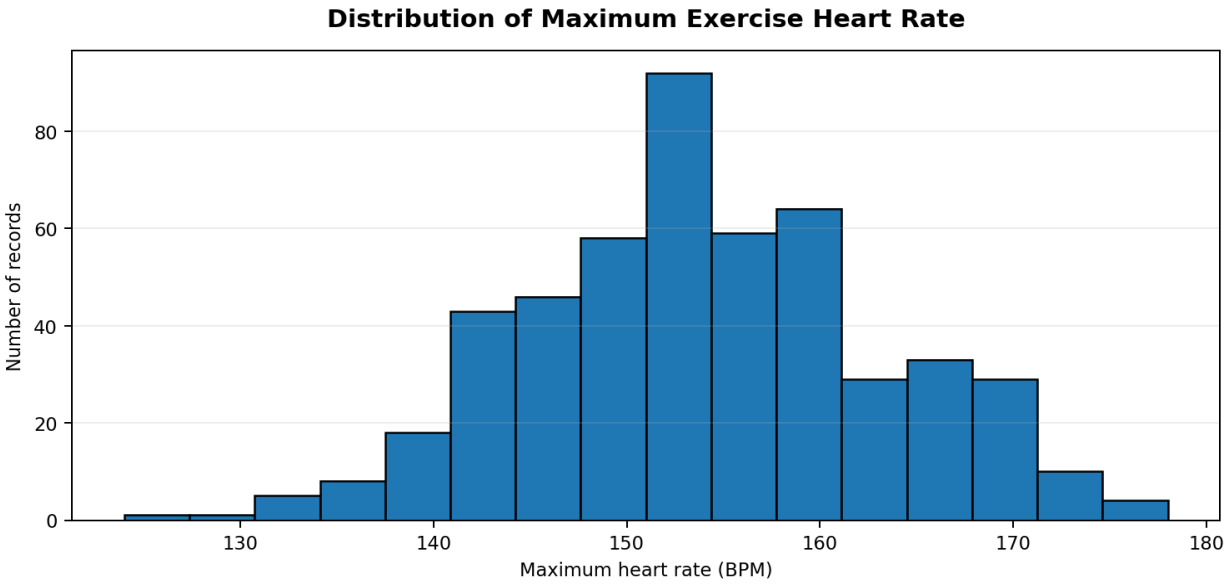


The strongest positive correlation among the original fields is between resting heart rate before and after (r = 0.97), indicating that the two measurements move closely together in this dataset. Exercise frequency and maximum exercise heart rate show a moderate negative linear correlation (r = –0.38). Sleep duration and maximum exercise heart rate show essentially no linear correlation (r ≈ 0.00). Correlation measures linear association and does not demonstrate causation.

Distribution of engineered heart-rate difference



Distribution of maximum exercise heart rate


Distribution of Maximum Exercise Heart Rate
Number of records
0
20
40
60
80
130
140
150
160
170
180
Maximum heart rate (BPM)

08 | Key Findings

Finding	Result / interpretation
Dataset size	500 rows and 6 original variables.
Data quality	0 missing cells and 0 exact duplicate rows in the supplied CSV.
Participant age	Mean age: 40.03 years (range: 18–60).
Sleep duration	Mean: 6.99 hours (range: 2.9–11.4).
Exercise frequency	Mean: 3.96 days per week (range: 1–7).
Resting HR before	Mean: 71.64 BPM.
Resting HR after	Mean: 64.68 BPM.
Maximum exercise HR	Mean: 153.89 BPM.
Derived HR difference	Mean: –6.96 BPM; median: –7 BPM.
Correlation pattern	Before/after resting HR: r = 0.97; exercise frequency / maximum exercise HR: r = –0.38.

Interpretation

The analysis demonstrates how a small set of structured variables can be explored using descriptive statistics, group summaries, feature engineering and visualization. The most noticeable statistical patterns are the close relationship between the two resting-heart-rate fields and the negative association between exercise frequency and maximum exercise heart rate. These patterns are specific to the supplied dataset and require further context before any broader interpretation.

09 | Limitations, Conclusion & Next Steps

Limitations

• The source dataset's collection process, sampling method and population characteristics were not provided with the CSV.
• The meaning and timing of the 'Resting Heart Rate After' measurement are unclear.
• Group averages can hide variation among individuals; some sleep and age groups contain fewer records than others.
• Correlation does not imply causation, and the analysis does not control for confounding variables.
• This project does not provide medical advice, diagnosis, risk assessment or individualized exercise recommendations.

Conclusion

HeartPulse AI provided a practical introduction to an end-to-end exploratory data analysis workflow. Using Python, Pandas, NumPy, Matplotlib and Seaborn, the project inspected a 500-row fitness dataset, checked data quality, summarized key measures, compared subgroups, engineered a heart-rate difference feature and visualized distributions and correlations. The work strengthened foundational skills in data handling, statistical summaries, charting and cautious interpretation.

Future development

1. Verify and document the original dataset source, collection method and measurement definitions.
2. Add richer variables such as activity type, exercise intensity, age-appropriate context and measurement timestamps.
3. Build an interactive dashboard for filtering and comparing groups.
4. After obtaining a suitable labelled dataset and defining a valid target, explore a machine-learning task with train/test separation and appropriate evaluation metrics.
5. Add reproducible notebook instructions and a clear README to the GitHub repository.

Personal learning project

This is an independently created learning and portfolio project, developed to build practical familiarity with data science concepts before formal training. It is not an academic submission or a production healthcare application.

References & project resources

• Python: https://docs.python.org/3/
• Pandas: https://pandas.pydata.org/docs/
• NumPy: https://numpy.org/doc/
• Matplotlib: https://matplotlib.org/stable/
• Seaborn: https://seaborn.pydata.org/
• Dataset: heart_rate_data.csv (add the verified original Kaggle URL and attribution before publishing).
• GitHub repository: add your repository URL here.